KAIZEN MATHS

Lesson starter

Column Vectors and Vector Geometry

Length

Name _____

S1

$\mathrm{v} = \begin{pmatrix} -4 \\ -2 \end{pmatrix}$

Find $|\mathrm{v}|$.

S2

$\mathrm{a} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$

$\mathrm{b} = \begin{pmatrix} -4 \\ -1 \end{pmatrix}$

Find the resultant a + b.

S3

$\mathrm{a} = \begin{pmatrix} -4 \\ -2 \end{pmatrix}$

$\mathrm{b} = \begin{pmatrix} -4 \\ -3 \end{pmatrix}$

Find $|\mathrm{a} + \mathrm{b}|$.

S4

$\mathrm{v} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$

Find $|\mathrm{v}|$.

Show your working

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Find $|\mathrm{a} + \mathrm{b}|$.

S4

$\mathrm{v} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$

Find $|\mathrm{v}|$.

GEO-7284-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Column Vectors and Vector Geometry

Length

Name

EA1

$$\mathrm{a} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

$$\mathrm{b} = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

Find the resultant a + b.

EA2

$$\mathrm{a} = \begin{pmatrix} -3 \\ 3 \end{pmatrix}$$

$$\mathrm{b} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$$

Find $|\mathrm{a} + \mathrm{b}|$.

KAIZEN MATHS

Exit ticket

Column Vectors and Vector Geometry

Length

Name

EB1

$$\mathrm{v} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$$

Find $|\mathrm{v}|$.

KAIZEN MATHS

Exit ticket

Column Vectors and Vector Geometry

Length

Name

EA1

$$\mathrm{a} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

$$\mathrm{b} = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

Find the resultant a + b.

EA2

$$\mathrm{a} = \begin{pmatrix} -3 \\ 3 \end{pmatrix}$$

$$\mathrm{b} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$$

Find $|\mathrm{a} + \mathrm{b}|$.

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Show your working

KAIZEN MATHS

Exit ticket

Column Vectors and Vector Geometry

Length

Name

EB1

$$\mathrm{v} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$$

Find $|\mathrm{v}|$.

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Show your working

Teacher answers & working

Column Vectors and Vector Geometry · Length

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S1

$\mathrm{v} = \begin{pmatrix} -4 \\ -2 \end{pmatrix}$

Find $|\mathrm{v}|$.

Answer:

$\sqrt{20} \approx 4.47$

Working:

$|\mathrm{v}| = \sqrt{(x^2 + y^2)}$

$|\mathrm{v}| = \sqrt{(-4^2 + -2^2)}$

$|\mathrm{v}| = \sqrt{20}$

$|\mathrm{v}| \approx 4.47$

S2

$\mathrm{a} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$

$\mathrm{b} = \begin{pmatrix} -4 \\ -1 \end{pmatrix}$

Find the resultant a + b.

Answer:

$\begin{pmatrix} -2 \\ 1 \end{pmatrix}$

Working:

Add matching components.

$\mathrm{a} + \mathrm{b} = \begin{pmatrix} 2 + -4 \\ 2 + -1 \end{pmatrix}$

$\mathrm{a} + \mathrm{b} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

Teacher answers & working

Column Vectors and Vector Geometry · Length

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S3

$\mathrm{a} = \begin{pmatrix} -4 \\ -2 \end{pmatrix}$

$\mathrm{b} = \begin{pmatrix} -4 \\ -3 \end{pmatrix}$

Find $|\mathrm{a} + \mathrm{b}|$.

Answer:

$\sqrt{89} \approx 9.43$

Working:

First find the resultant.

$\mathrm{a} + \mathrm{b} = \begin{pmatrix} -4 + -4 \\ -2 + -3 \end{pmatrix} = \begin{pmatrix} -8 \\ -5 \end{pmatrix}$

$|\mathrm{a} + \mathrm{b}| = \sqrt{(-8^2 + -5^2)} = \sqrt{89}$

$|\mathrm{a} + \mathrm{b}| \approx 9.43$

S4

$\mathrm{v} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$

Find $|\mathrm{v}|$.

Answer:

$\sqrt{13} \approx 3.61$

Working:

$|\mathrm{v}| = \sqrt{(\mathrm{x}^2 + \mathrm{y}^2)}$

$|\mathrm{v}| = \sqrt{(-2^2 + 3^2)}$

$|\mathrm{v}| = \sqrt{13}$

$|\mathrm{v}| \approx 3.61$

Teacher answers & working

Column Vectors and Vector Geometry · Length

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EA1

$\mathrm{a} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$

$\mathrm{b} = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$

Find the resultant a + b.

Answer:

$\begin{pmatrix} 4 \\ -4 \end{pmatrix}$

Working:

Add matching components.

$\mathrm{a} + \mathrm{b} = \begin{pmatrix} 3+1 \\ -1+-3 \end{pmatrix}$

$\mathrm{a} + \mathrm{b} = \begin{pmatrix} 4 \\ -4 \end{pmatrix}$

EA2

$\mathrm{a} = \begin{pmatrix} -3 \\ 3 \end{pmatrix}$

$\mathrm{b} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$

Find $|\mathrm{a} + \mathrm{b}|$.

Answer:

1

Working:

First find the resultant.

$\mathrm{a} + \mathrm{b} = \begin{pmatrix} -3+4 \\ 3+-3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$|\mathrm{a} + \mathrm{b}| = \sqrt{(1^2 + 0^2)} = \sqrt{1}$

$|\mathrm{a} + \mathrm{b}| \approx 1.00$

Teacher answers & working

Column Vectors and Vector Geometry · Length

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EB1

$\mathrm{v} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$

Find $|\mathrm{v}|$.

Answer:

$\sqrt{10} \approx 3.16$

Working:

$|\mathrm{v}| = \sqrt{(x^2 + y^2)}$

$|\mathrm{v}| = \sqrt{(-1^2 + 3^2)}$

$|\mathrm{v}| = \sqrt{10}$

$|\mathrm{v}| \approx 3.16$